

Data Science Curriculum

MATHS Module

Summary: This section covers foundational math skills including linear algebra, probability, and calculus essential for data analysis.

Day	Topic	Resources and Activities	Objective/Outcome
	FUNDAMENTALS OF MATHEMATICS	https://www.geeksforgeeks.org/blogs/maths-for-data-science/	
1.0	Linear Algebra	Scalars, Vectors, Matrices	Gain the ability to model, analyze, and solve real-world problems using vectors, matrices, and linear systems, essential for AI and data science.
2.0	probability&statistics	Bayes Theorem, statistical tests	Develop the ability to analyze data, measure uncertainty, and make data-driven decisions using statistical methods and probabilistic reasoning.
3.0	calculus	Diffrentiation, Partial derrivatives, integrals	Understand how things change by applying derivatives and integrals to model, analyze, and optimize continuous systems in science and engineering.

ANALYTICS Module

Summary: This section introduces Python's Pandas library for data manipulation, covering loading, filtering, grouping, and cleaning data.

Da y	Topic	Resource Link	Outcome
1	Python Pandas – Introduction	https://www.kaggle.com/learn/pandas	Load and explore datasets using

			DataFrames
	Pandas – Selecting Data	https://www.kaggle.com/learn/pandas	Select data using loc[], iloc[]
	Pandas – Filtering & Sorting	https://www.kaggle.com/learn/pandas	Apply filters, sort rows/columns
2	Pandas – Aggregation & GroupBy	https://www.kaggle.com/learn/pandas	Summarize and group data effectively
	Pandas – Missing Values	https://www.kaggle.com/learn/pandas	Handle nulls using fillna(), dropna()
	Pandas – Merging & Joining	https://www.kaggle.com/learn/pandas	Combine multiple datasets
3	Pandas – Time & Categorical Data	https://www.kaggle.com/learn/pandas	Handle datetime and categorical data
	NumPy – Basics	https://www.kaggle.com/code/legendadnan/numpy-tutorial-for-beginners-data-science	Learn array creation and indexing
	SciPy – Scientific Computing	https://www.kaggle.com/code/imdevskp/scipy	Perform math and scientific computations
4	Matplotlib – Plotting Basics	https://www.kaggle.com/code/prashant111/matplotlib-tutorial-for-beginners	Create visualizations : line, bar, scatter
	Matplotlib – Customization	https://www.kaggle.com/code/prashant111/matplotlib-tutorial-for-beginners	Customize charts: titles, colors, labels
	Seaborn – Dataset Overview	https://www.kaggle.com/datasets/abdoomoh/all-seaborn-built-in-datasets	Load and visualize Seaborn datasets
	Seaborn – Styling & Themes	https://www.kaggle.com/learn/data-visualization	Format charts with themes and palettes
5	Data Cleaning in Pandas	https://www.kaggle.com/learn/data-cleaning	Handle nulls, incorrect types, outliers
	Feature Engineering	https://www.kaggle.com/learn/feature-engineering	Create meaningful new features
	Scikit-learn – Basics	https://www.kaggle.com/code/nicolausboby/introduction-to-scikit-learn	Train simple ML models
6	R – Introduction & Basics	https://www.w3schools.com/r/	Learn R syntax, variables & data types

	R – Data Structures	https://www.w3schools.com/r/	Work with vectors, lists, matrices, data frames
7	R – Data Analysis Basics	https://www.w3schools.com/r/	Filter, select, and summarize data
	R – Data Visualization	https://www.w3schools.com/r/	Plot charts using base R and ggplot2
8	SQL – SELECT & WHERE	https://www.kaggle.com/learn/intro-to-sql	Query data using SELECT and WHERE
	SQL – Aggregation Functions	https://www.kaggle.com/learn/intro-to-sql	Use SUM, COUNT, AVG, etc.
	SQL – GROUP BY & HAVING	https://www.kaggle.com/learn/intro-to-sql	Group and filter grouped data
9	SQL – JOIN Operations	https://www.kaggle.com/learn/intro-to-sql	Combine tables using JOIN
	SQL – Nested Queries	https://www.kaggle.com/learn/intro-to-sql	Use subqueries and complex filtering
10	Excel – Sort, Filter, Formulas	https://support.microsoft.com/en-us/training	Organize and calculate data in Excel
	Excel – Charts & Formatting	https://support.microsoft.com/en-us/training	Create and format data charts
	Excel – Pivot Tables	https://support.microsoft.com/en-us/training	Summarize large datasets using PivotTables
11	Excel – IF, VLOOKUP, Automation	https://support.microsoft.com/en-us/training	Use functions for data logic and automation
	Excel – Dashboard Creation	https://support.microsoft.com/en-us/training	Build interactive dashboards using slicers, charts, layout
12	Time series Analysis-intro	https://www.analyticsvidhya.com/blog/2021/10/a-comprehensive-guide-to-time-series-analysis/	To know the terms and basic usage of TSA
	Time series Analysis-working	https://youtu.be/MmC4b7gPY0Q?si=744MGOKsT2NsnAAV	How python is used for TSA

VISUALIZATION Module

Summary: Focused on tools like Tableau and Power BI, this section helps create interactive dashboards and visualize data effectively.

Day	Topic	Resources and Activities	Objective/Outcome
	VISUALIZATION		
1.0	Tableau	https://youtu.be/olw8xJ1Fy3w?si=qRhWdLEXSzAN7jG	Create interactive dashboards and data visualizations to communicate insights clearly.
2.0	Tableau	https://youtu.be/TPMIZxRRaBQ?si=MdUd0CHAo43K3B1e	Connect, transform, and analyze diverse datasets for effective business intelligence reporting.
3.0	PowerBI	https://youtu.be/3u7MQz1EyPY?si=Ye_Ehw1fzoXJXXKA	Design dynamic dashboards and reports to visualize and share data insights interactively.
4.0	PowerBI	https://youtu.be/mmxVCFceQgU?si=w9hoRWFfwLCPwpP1	Integrate and transform data from multiple sources for real-time analytics and decision-making.

WEB SCRAP Module

Summary: Teaches web scraping using BeautifulSoup and Selenium, enabling automation and data extraction from websites.

Day	Topic	Resources and Activities	Objective/Outcome
	WEB SCRAPING		
1.0	Python and BeautifulSoup	https://youtu.be/kJyOp12PgP4?si=8WcP6h9aBbVWkt3L	Learn how web pages are structured using HTML tags and attributes.

2.0	Python and BeautifulSoup	https://www.youtube.com/live/rIR0f4zZKvc?si=OxxkvfYbw pULld9n	Use BeautifulSoup to parse HTML/XML files and navigate the document tree.
3.0	Selenium	https://youtu.be/j7VZsCCnptM?si=B7BhDQxsHA-rRJQh	Automate interactions with web browsers (Chrome, Firefox, etc.)—like a human user.
4.0	Selenium	https://www.scaler.com/topics/data-science/web-scraping/	Capture screenshots and manage browser pop-ups or alerts.
5.0	Selenium	Scrap datas from web using Selenium	

BIG DATA Module

Summary: Introduces the Hadoop ecosystem, covering fundamentals like HDFS, MapReduce, and handling large-scale datasets.

Day	Topic	Resources and Activities	Objective/Outcome
	BIG DATA	https://youtu.be/1vbXmCrkT3Y?si=tPWDEo5CsFpbH-zv	
1.0	Hadoop Fundamentals	Introduction to Hadoop, HDFS Overview	Understand Hadoop ecosystem and HDFS architecture
2.0	Hadoop Fundamentals	HDFS Architecture & Commands	Learn HDFS structure and file operations
3.0	Hadoop Fundamentals	MapReduce Concepts	Learn MapReduce workflow and components
4.0	Hadoop Fundamentals	MapReduce Job Lifecycle	Run a sample MapReduce job in dev mode
5.0	Hive & Pig	Hive Data Modeling & SQL Queries	Write Hive DDL/DML queries
6.0	Hive & Pig	Pig Latin Basics & Data Flows	Construct Pig scripts for ETL
7.0	YARN & HBase	YARN Components & HBase Data Model	Understand resource manager and HBase tables
		https://youtu.be/94w6hPk7nkM?si=st0vW0Z0CyFp6D5C	
8.0	Spark Intro	Spark Architecture & Jobs	Run basic Spark job demonstrating transformations/actions
9.0	Hadoop Mini Project	Build MapReduce WordCount	Apply Hadoop skills to process sample

			data
10.0	PySpark Essentials	RDD Creation & Transformations	Create and manipulate RDDs programmatically
11.0	PySpark Essentials	DataFrame API & Schemas	Read/write DataFrames, understand schema
12.0	PySpark Essentials	Spark SQL Queries & Optimization	Write SQL queries and optimize execution
13.0	PySpark Essentials	Combined RDD/DataFrame Exercises	Solidify foundational PySpark skills
14.0	Advanced PySpark	Complex Transformations & Joins	Build advanced data processing pipelines
15.0	Advanced PySpark	MLlib – Pipelines, Estimators, Transformers	Build a basic machine learning pipeline
16.0	PySpark Mini Project	ETL Pipeline: CSV → Parquet + Parquet Reporting	Implement full ETL flow and verify output
		https://youtu.be/c2M-rlkkT5o?si=prO_ZV0xTkXFoewU	
17.0	MongoDB Setup & CRUD	MongoDB Schema Design & CRUD Ops	Perform create, read, update, delete operations
18.0	Aggregation & Indexing	Aggregation Framework & Index Types	Write aggregation queries and use indexes
19.0	MongoDB Integration	PySpark–MongoDB Connector Usage	Exchange data between Spark and MongoDB